PRACTICE ANSWERS

Unit 1: Completing the Square

1. $x = -3,\ x = -5$
2. $x = 3 \pm \sqrt{7}$
3. $x = \frac{-9 \pm \sqrt{73}}{2}$
4. $x = 1 \pm \frac{\sqrt{3}}{2}$
5. $\frac{1}{2} \pm i$

Unit 2: Radical Manipulation

1. $2\sqrt{3}$
2. $2\sqrt{5}$
3. $3\sqrt{3}$
4. $5\sqrt{2}$
5. $6\sqrt{2}$
6. $14\sqrt{2}$
7. $-15\sqrt{6}$
8. $-18\sqrt{2}$
9. $60\sqrt{2}$

Unit 3: Triangle Congruence

1.
   a. SAS
   b. $\Delta FIG \cong \Delta HIG$
2.
   a. SSS
   b. $\Delta QPN \cong \Delta OPN$
3.
   a. ASA
   b. $\Delta ABC \cong \Delta DEC$
4.
   a. ASA
   b. $\Delta RSU \cong \Delta TSU$
5.
   a. AAS
   b. $\Delta KJM \cong \Delta MLK$
6.
   a. SAS
   b. $\Delta OPN \cong \Delta QPN$
7.
   a. SAA
   b. $\Delta ABC \cong \Delta DEC$
8.
   a. ASA
   b. $\Delta FGI \cong \Delta HGI$
9.
   a. SSS
   b. $\Delta KJM \cong \Delta MLK$
10.
   a. ASA
   b. $\Delta NOP \cong \Delta NQP$
11. DOESN'T NECESSARILY WORK
   a. ASS
12.
   a. SSS
   b. $\Delta RSU \cong \Delta TSU$

Unit 4: Angle Theorem

1. 75°
2. 105°

1. 123°
2. 123°

1. 66°
2. 116°

Unit 5: Trigonometry

$x \approx 5.4'$

$x \approx 53.6'$

$x \approx 6.2'$

Unit 6: Two Column Proofs

4.

STATEMENTS	REASONS
1. $\overline{AC} \cong \overline{DB}$	1. Given
2. $\overline{AB} \perp \overline{BC},\ \overline{CD} \perp \overline{BC}$	2. Given
3. $\angle C \cong \angle B$	3. Definition of $\perp$ lines
4. $\triangle ACB$ and $\triangle DBC$ are right triangles.	4. Definition of a right triangle

Statements	Reasons
1. $\angle A$ and $\angle C$ are complementary $\angle B$ and $\angle C$ are complementary	1. Given
2. $\angle A + \angle C = 90°$ $\angle B + \angle C = 90°$	2. Definition of Complementary Angles
3. $\angle A + \angle C = \angle B + \angle C$	3. Transitive Property
4. $\angle A = \angle B$	4. Subtraction Property
5. $\angle A \cong \angle B$	5. Definition of Congruent Angles

Unit 7: Statistics

1.

a. $\frac{4}{24}$

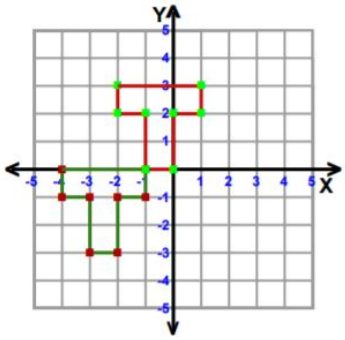
b. $\frac{21}{24}$

c. $\frac{4}{24} = \frac{1}{6}$

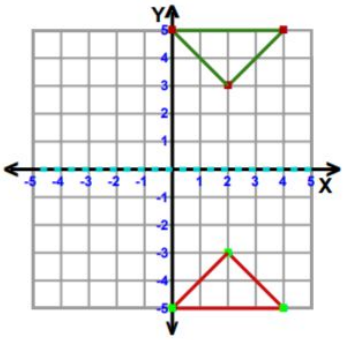
d. $\frac{21}{24} > \frac{1}{6}$

Unit 8: Transformations

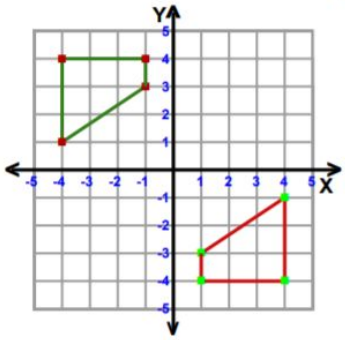
1) Translation: 2 right and 3 up



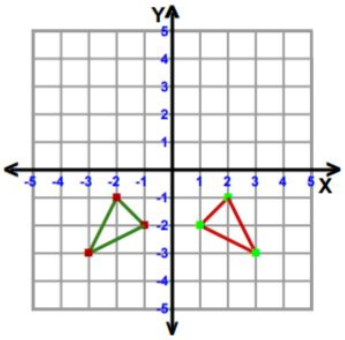
2) Reflection: Across the x-axis



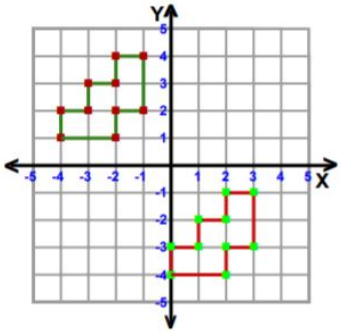
3) Rotation: 180° about the origin



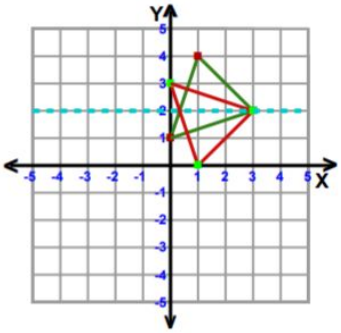
4) Rotation: 90° ccw about the origin



5) Translation: 4 right and 5 down



6) Reflection: Across the line y = 2



Unit 9: Projectile Motion

1. $h(t) = -16t^2 + 40t + 50$
   a. $t = 1.25$ *seconds*
   b. 75 ft
2. 80 ft
3. 96 ft/sec